

Internet of Things

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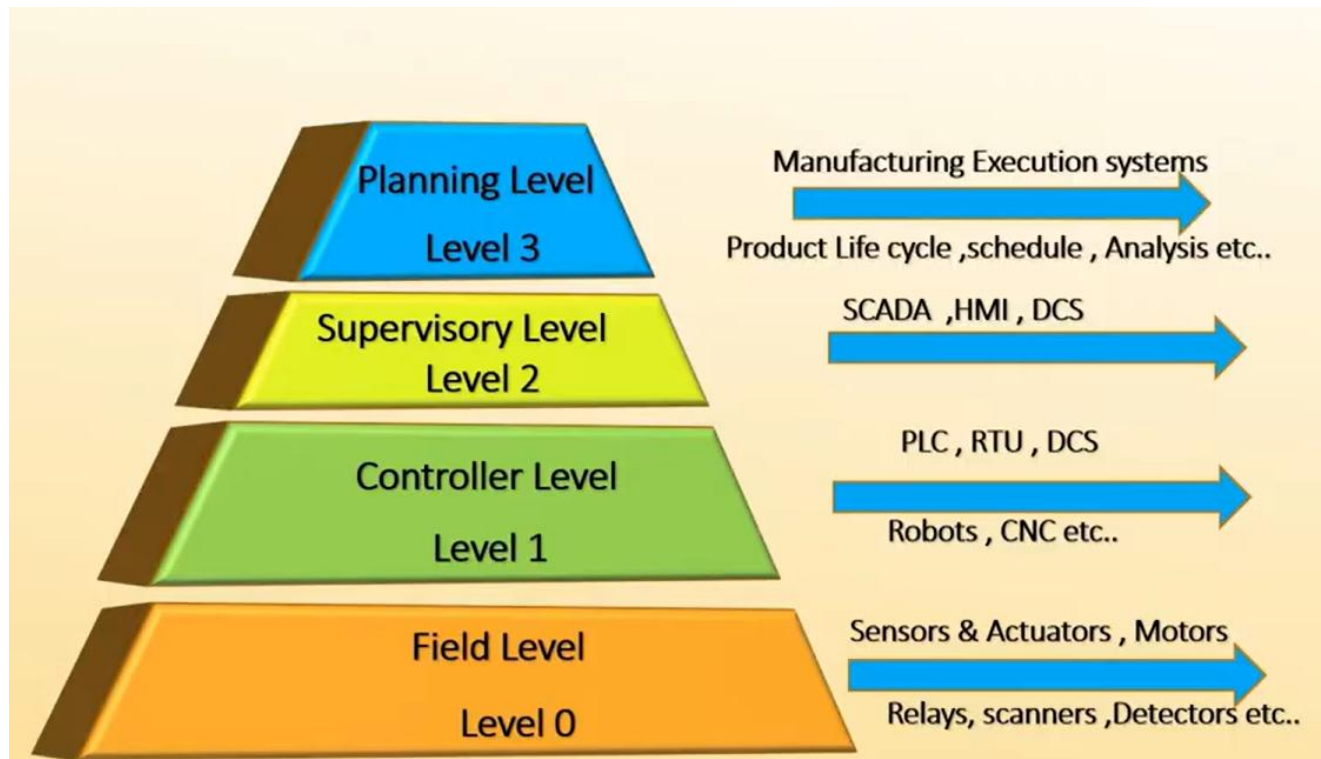
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Outline

- ▶ Introduction to Industrial IoT
- ▶ Evolution of the Factory
- ▶ Key Components of Industrial IoT
- ▶ Applications of Industrial IoT
- ▶ Future improvement in Industrial IoT
- ▶ Practice code Industrial IoT

Evolution of the Factory

- ▶ **The Paradigm Shift:** Moving from the traditional **Automation Pyramid (ISA-95)** to a decentralized, flat network.



Evolution of the Factory

- ▶ The transition from the rigid, hierarchical ISA-95 Automation Pyramid to a decentralized, flat architecture represents the fundamental shift from Industry 3.0 to Industry 4.0.
- ▶ In the traditional model, data flows linearly through layers
(Sensing → Control → SCADA → MES → ERP), creating bottlenecks and "data silos." The modern shift replaces this with a Unified Namespace (UNS) or an industrial network fabric where every asset can communicate directly.
- ▶ **SCADA (Supervisory Control and Data Acquisition):** Focuses on the machinery. It gathers real-time data from sensors and PLCs to monitor, control, and visualize factory processes.

Evolution of the Factory

- ▶ MES (Manufacturing Execution System): Functions as the bridge between production and business. It translates ERP orders into action, tracking, tracing, and improving production efficiency.
- ▶ ERP (Enterprise Resource Planning): Handles business-level operations. It manages finances, procurement, scheduling, and supply chain, providing the overall production goals to the MES

Evolution of the Factory

- ▶ Focus: SCADA controls machines; MES manages the process; ERP manages the business.
- ▶ Timeframe: SCADA operates in milliseconds; MES works in real-time/shifts; ERP operates in days/weeks.
- ▶ Users: Operators/engineers (SCADA), production supervisors (MES), managers/executives (ERP).

Evolution of the Factory (OT vs IT)

- ▶ **Operational Technology (OT):** Hard-real-time systems, PLCs (Programmable Logic Controllers), and SCADA. Priority: Availability and Safety.
- ▶ **Information Technology (IT):** Data-driven, cloud-based, and interconnected. Priority: Confidentiality and Data Integrity.
- ▶ **Core Drivers:** Predictive maintenance, asset tracking, and energy efficiency.

Introduction to Industrial IoT

- ▶ Industrial IoT (IIoT) signifies the application of Internet of Things (IoT) technology within industrial environments. This involves the intricate networking of a multitude of industrial assets, encompassing machinery, devices, sensors, and entire systems, to establish a cohesive network capable of real-time data collection, seamless exchange, and sophisticated analysis. The central aim of IIoT is to foster intelligent industrial operations, ultimately leading to substantial gains in both efficiency and overall productivity.



Introduction of IIoT

- ▶ Industrial IoT is an ecosystem of devices, sensors, applications, and associated networking equipment that work together to collect, monitor, and analyze data from industrial operations. Analysis of such data helps increase visibility and enhances troubleshooting and maintenance capabilities. It can also increase efficiencies, reduce costs, and improve safety and security.



Introduction to Industrial IoT

- ▶ By facilitating the immediate acquisition and in-depth analysis of operational data, **IIoT empowers organizations to achieve a profound level of optimization across their operations.** This includes the ability to accurately predict potential equipment failures, consistently enhance the quality of manufactured products, and significantly improve the safety standards for personnel.

The IIoT Architecture Stack

- ▶ A robust IIoT system is typically divided into four functional layers:
- ▶ **Perception Layer (Edge):** Sensors (vibration, temperature, pressure) and actuators.
- ▶ **Connectivity Layer:** How the "Things" talk.
- ▶ **Protocols:** MQTT (Lightweight, Pub/Sub), OPC UA (Industry standard for interoperability), and Modbus.
- ▶ **Platform/Middleware Layer:** Data ingestion, device management, and "Digital Twin" creation—a virtual representation of physical assets.
- ▶ **Application Layer:** The "Business Logic." ERP integration, dashboards, and remote monitoring.

Connectivity and Edge Computing

- ▶ Why Edge? In industrial settings, sending all data to the cloud is often impossible due to:
- ▶ Latency: Millisecond responses required for safety.
- ▶ Bandwidth: Thousands of sensors generating high-frequency data.
- ▶ Security: Keeping sensitive process data on-premises.
- ▶ Networking Standards: The role of 5G in the factory (low latency, high density) and TSN (Time-Sensitive Networking) for deterministic Ethernet.

Advanced Use Cases: AI & Digital Twins

- ▶ **Predictive Maintenance (PdM):** Using vibration and thermal data to predict a motor failure before it happens, moving from "fixed-interval" to "condition-based" maintenance.
- ▶ **Digital Twins:** Running simulations on a virtual model of a turbine to optimize performance without risking the physical hardware.
- ▶ **Green IoT:** Real-time monitoring of carbon footprints and energy optimization in manufacturing lines.

Application of Industrial IoT

- ▶ IIoT enables companies to gather and meticulously analyze vast quantities of data, transforming this raw information into actionable insights that directly address operational challenges and expedite the process of making critical decisions.

Challenges: Security & Interoperability

- ▶ The Security Gap: Legacy industrial equipment was never designed to be online.
- ▶ Threat Vectors: Ransomware in manufacturing, unauthorized access to control loops.
- ▶ Interoperability: The "Mabel's Soup" of proprietary protocols vs. open standards like OPC UA.

Differentiating IIoT from general IoT concepts

- ▶ While both IIoT and the broader concept of IoT revolve around the principle of interconnected devices, their primary distinction lies in the specific domains of their application.
- ▶ This enables direct and automated communication and data exchange between various industrial assets, such as robots on an assembly line or sensors monitoring the condition of a critical piece of equipment.

Differentiating IIoT from general IoT concepts

- ▶ IIoT systems typically encompass significantly larger networks with a far greater quantity of data being generated, transmitted, and processed compared to typical consumer-focused IoT deployments.
- ▶ Given the **mission-critical nature** of industrial operations and the potential for significant safety and economic consequences resulting from disruptions, cybersecurity requirements in IIoT are considerably more **stringent** than those in consumer IoT. The protection of industrial systems from cyber threats demands the implementation of advanced and highly resilient cybersecurity protocols and measures.

Differentiating IIoT from general IoT concepts

- ▶ The integration of IIoT with existing legacy industrial systems, which often rely on proprietary communication protocols and have extended operational lifespans, introduces a unique set of challenges that are not typically a concern in the realm of consumer IoT. Successfully addressing these challenges requires meticulous planning and the adoption of specialized solutions to ensure seamless interoperability and robust security.

Highlighting the key benefits and potential impact of IIoT across industries

- ▶ One of the most prominent advantages is the significant potential to boost overall efficiency within operations. This is achieved through the implementation of automation technologies and the continuous, real-time monitoring of both equipment and industrial processes. The availability of immediate data empowers organizations to make informed, data-driven decisions and to implement remote monitoring capabilities, ultimately leading to the optimized allocation and utilization of valuable resources.

key benefits and potential impact of IIoT across industries

- ▶ IIoT also plays a crucial role in achieving a substantial increase in production output. By establishing interconnected manufacturing processes, IIoT enhances the overall usage and performance of industrial equipment. Maximizing the effectiveness of machinery and liberating human employees from performing repetitive tasks directly contributes to an expansion of the organization's total production capacity.

key benefits and potential impact of IIoT across industries

- ▶ The integration of automation into industrial operations through IIoT leads to a notable reduction in errors that are typically associated with human involvement. This results in the production of higher-quality products with fewer defects. Consequently, businesses experience improved levels of customer satisfaction and a stronger brand reputation within the marketplace.

key benefits and potential impact of IIoT across industries

- ▶ A particularly significant benefit of IIoT is its capacity to enable the implementation of sophisticated predictive maintenance strategies. By utilizing IIoT sensors to meticulously analyze production data for recurring patterns and early indicators of potential issues, organizations can proactively address maintenance needs.

Core Components of IIoT Systems

- ▶ **Smart sensors** serve as the fundamental building blocks of Industrial IoT (IIoT) systems. These sophisticated devices are equipped with sensing elements, embedded processing capabilities, and communication interfaces, enabling them to gather real-time data from the physical industrial environment and convert this information into digital signals. E.g. RFID
- ▶ Actuators, in contrast, are responsible for translating digital commands received from the IIoT system into physical actions within the industrial setting.

Role of edge devices and gateways in data processing and connectivity

- ▶ Edge devices are strategically deployed within the Industrial IoT (IIoT) architecture to perform data processing tasks in close proximity to the data sources. This localized processing is essential for minimizing latency, which is particularly critical for real-time applications such as autonomous control systems and safety-critical operations.

Key Considerations for Real-World Industrial IoT

- ▶ **Scalability:** Real-world Industrial IoT systems often involve a large number of devices generating massive amounts of data. The architecture and code need to be designed to handle this scale.
- ▶ **Reliability and Fault Tolerance:** Industrial environments can be harsh, and devices or network connections can fail. The system needs to be robust and able to handle failures gracefully. This often involves redundancy, error handling, and fault detection mechanisms.

Key Considerations for Real-World Industrial IoT

- ▶ **Security:** Security is paramount in Industrial IoT, as these systems often control critical infrastructure. The code and system design must address security concerns such as device authentication, data encryption, and access control.
- ▶ **Real-time Processing:** Many Industrial IoT applications require real-time or near-real-time processing of data to enable timely decision-making and control.

Key Considerations for Real-World Industrial IoT

- ▶ **Communication Protocols:** Industrial IoT devices use a variety of communication protocols, such as MQTT, OPC-UA, and Modbus. The code needs to be able to communicate with devices using these protocols.
- ▶ **Data Storage:** Efficient and scalable data storage is essential for Industrial IoT. Time-series databases like InfluxDB or TimescaleDB are often used to store sensor data.
- ▶ **Cloud Integration:** Many Industrial IoT systems leverage cloud platforms for data storage, processing, analytics, and visualization.

Practice code

- ▶ Modify the `analyze_data` function to include a check for this new low-temperature threshold.
- ▶ When the `temperature` falls below `65°C`, print a message to the console with a timestamp indicating a "Low Temperature Warning!" and the current temperature.

Reference

- ▶ [https://www.cisco.com/c/en/us/solutions/internet-of-things/what-is-industrial-
iot.html#:~:text=Industrial%20IoT%20is%20an%20ecosystem,analyze%20data%20from%20industrial%20operations](https://www.cisco.com/c/en/us/solutions/internet-of-things/what-is-industrial-
iot.html#:~:text=Industrial%20IoT%20is%20an%20ecosystem,analyze%20data%20from%20industrial%20operations).

Question



Thank you

